DATA STRUCTURES ASSIGNMENT 3 – CO3

Submission Evidence Appendix

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Repository evidence included

Requirement (page 22)	Evidence location	Status
C source code for all questions	src/q1_bst.c … src/q7_dictionary.c	Included
Input files	inputs/	Included
Executed output	outputs/*.txt	Included
Execution screenshots	screenshots/*_execution.png	Included
Tree diagrams	diagrams/*.png	Included
Predictions and analysis	predictions_and_analysis.md	Included
README / build instructions	README.md	Included
GitHub repository URL	Must be created and pasted by student	Pending

Execution evidence summary

Q1 – BST: Search comparisons, traversals, deletion cases, height and a 2,000-key insertion experiment.

Q2 – AVL: Rotations, final tree, balance verification and sorted insertion of 100,000 keys with CPU timings.

Q3 – B-tree: Minimum-degree-2 insertions, node splits, sorted traversal and search node-read counts.

Q4 – Red-black tree: Insertion sequence and red-black invariant verification after each insertion.

Q5 – Splay tree: Repeated accesses to 3 and 9 showing the accessed key becoming the root.

Q6 – Trie: Prefix counts, autocomplete, exact search and deletion test.

Q7 – Dictionary: Exact lookup, alphabetical range and autocomplete using a small sample fixture.

Important scope and verification notes

• The attached C files were compiled and run in the preparation environment. Output text files are captured stdout; terminal-style PNGs are visual evidence of those captured outputs.

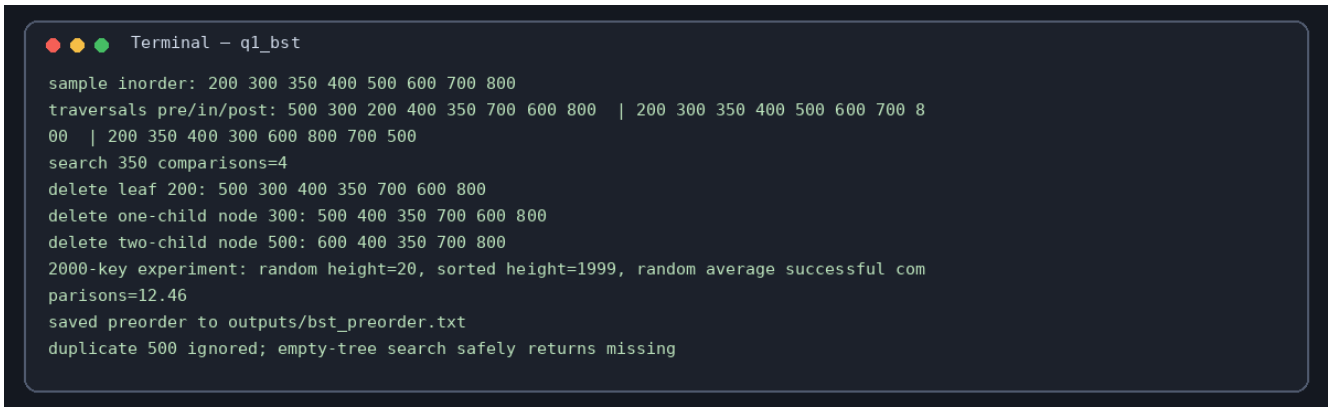
• Timing values depend on the computer, compiler and load. Re-run the programs on the submission computer if faculty expects locally measured timings.

• The current implementations are assignment-focused demonstrations. B-tree deletion, red-black deletion, splay deletion, complete 5,000/50,000-contact trie benchmarking and a production-scale 200,000-word dictionary are not claimed as implemented. See README.md and SUBMISSION_CHECKLIST.md.

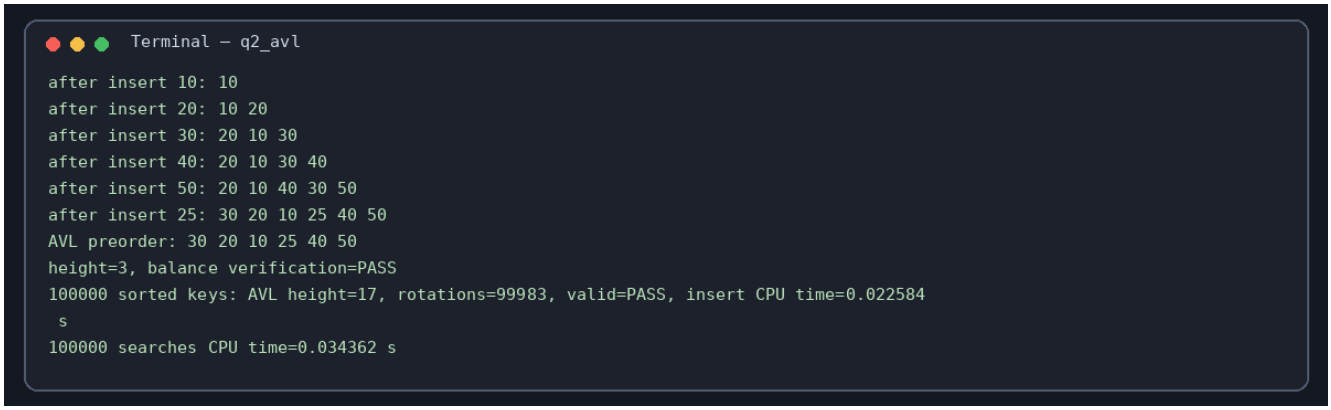
• A real GitHub repository cannot be represented by a placeholder URL. Create the repository, push this folder using the README commands, and submit its actual URL.

Sample execution screenshots

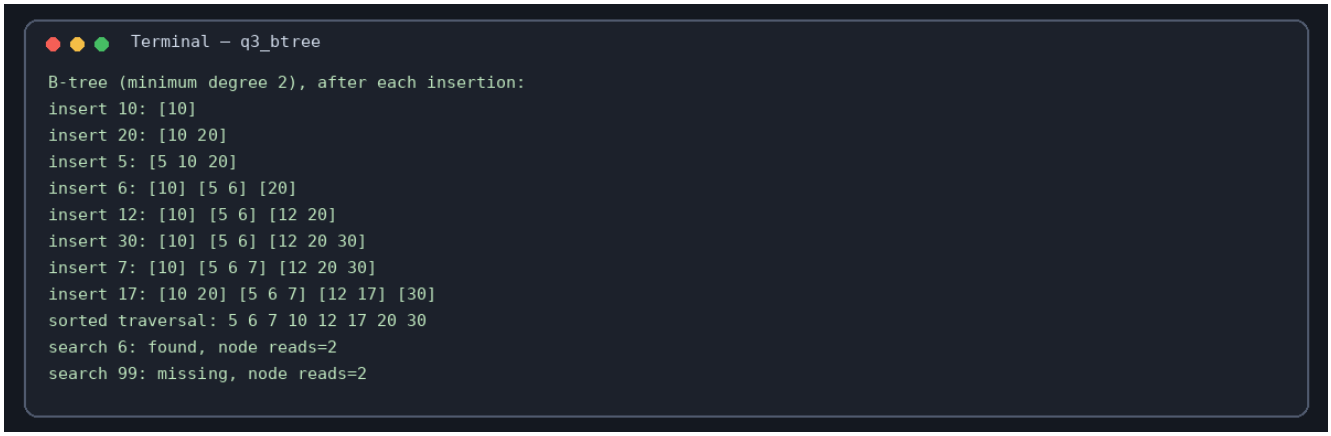
Q1 BST



Q2 AVL



Q3 BTREE



Q4 RED BLACK

```
Terminal — q4_red_black

Red-black tree after each insertion (inorder key/color):
insert 10: 10(B) | properties PASS
insert 20: 10(B) 20(R) | properties PASS
insert 30: 10(R) 20(B) 30(R) | properties PASS
insert 15: 10(B) 15(R) 20(B) 30(B) | properties PASS
insert 25: 10(B) 15(R) 20(B) 25(R) 30(B) | properties PASS
insert 5: 5(R) 10(B) 15(R) 20(B) 25(R) 30(B) | properties PASS
Minimum process key: 5
```

Q5 SPLAY

```
Terminal — q5_splay

Initial preorder: 9 8 7 6 5 4 3 2 1
access 3 -> root 3, preorder: 3 2 1 8 6 4 5 7 9
access 9 -> root 9, preorder: 9 8 3 2 1 6 4 5 7
access 3 -> root 3, preorder: 3 2 1 8 6 4 5 7 9
access 9 -> root 9, preorder: 9 8 3 2 1 6 4 5 7
access 3 -> root 3, preorder: 3 2 1 8 6 4 5 7 9
access 9 -> root 9, preorder: 9 8 3 2 1 6 4 5 7
```

Q6 TRIE

```
Terminal — q6_trie

a:5 an:4 anu:2 b:1 z:0
autocomplete 'an': anand anitha anu anusuya
search anu: found
delete anu: success; prefix anu count=1
```

Q7 DICTIONARY

```
Terminal — q7_dictionary

exact 'band': found
exact 'bad': missing
alphabetical range [banana..cat]: banana band bandit cat
autocomplete 'ba': banana band bandit
verified dictionary words=10
```